

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Description about the assessment tool:

Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.
2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every

28
50
56

window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP
throughput
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7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 3 - Exemplary	Level 2 - Proficient	Level 1 - Developing	Level 0 - Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Presentation	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

The Transport Layer 4 is responsible for dividing large data into smaller segments, providing end-to-end communication, error recovery and flow control

Given:

File Size = 150 MB, Bandwidth = 50 Mbps, Segment Size = 1500 Bytes

Data Link overhead = 40 Bytes / frame

First we need to convert file size

$$150 \text{ MB} = 150 \times 1,000,000 = 150,000,000 \text{ Bytes}$$

$$\text{Segments} = \frac{150,000,000}{1500} = 100,000$$

$\therefore$ Number of frames and segments = 100,000

overhead per frame = 40 Bytes

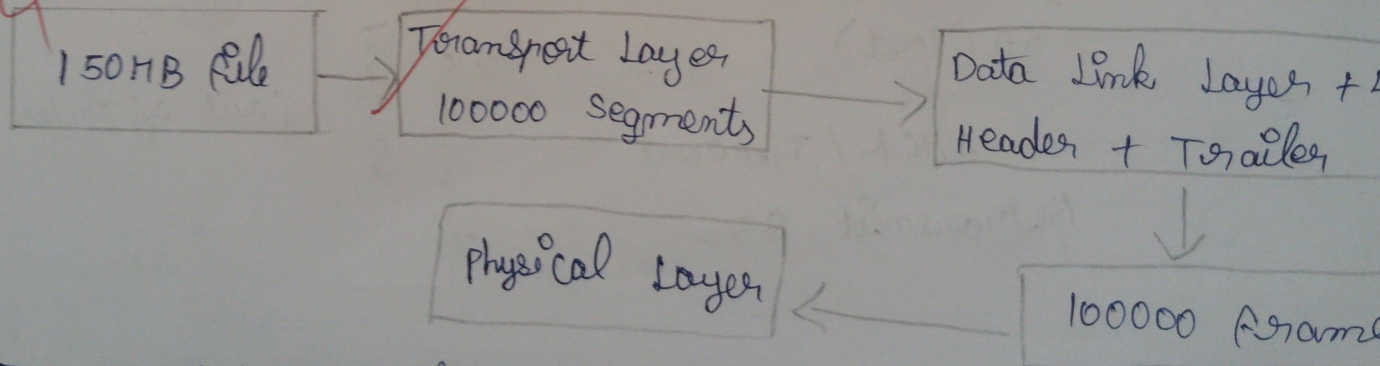
$$100,000 \times 40 = 4,000,000 \text{ Bytes}$$

So the Total overhead is 4 MB

$$150 \text{ MB} = 1200 \text{ Mb}$$

$$\frac{1200}{50} = 24 \text{ s}$$

Therefore the Transmission Time is 24 Seconds



* Transport Layer splits the file into segments, uses acknowledgments and retransmissions for reliability

* Data Link Layer converts segments into frames and retransmits damaged

ed by transport layer. It allows multiple frames to be transmitted before receiving acknowledgements

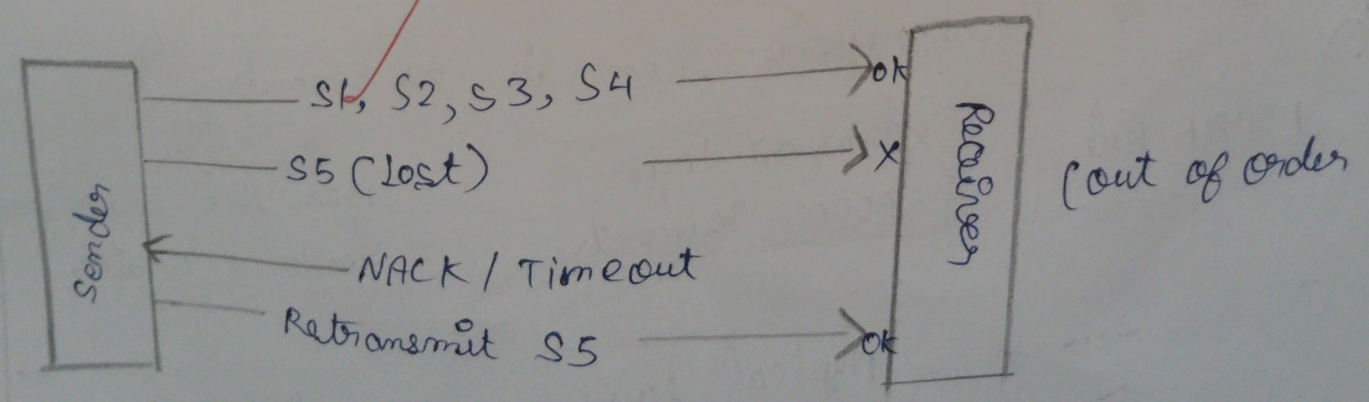
Given:
Window Size = 6 frames
Frame Size = 1024 Bytes
Total frames = 48

calculate number of windows

$$\frac{48}{6} = 8$$

The windows Required is 8
A frame is lost in every window
before Retransmission = 8 frames

Flow control Improves communication efficiency by matching the Sender's Transmission rate with the receiver's Processing capability



The Network Layer 3 is responsible for logical addressing, routing and packet forwarding. When a packet exceeds the Maximum Transmission Unit, the Network Layer fragments it into smaller packets.

Given:

IP Packet = 12,000 Bytes, MTU = 1500 Bytes, IP Header = 20 Bytes

Payload per fragment

$$1500 - 20 = 1480 \text{ Bytes}$$

Payload = 1480 Bytes

Number of fragments

$$\frac{12000}{1480} = 8.1$$

Rounding up as 9

Each fragment has a 20-byte header

$$9 \times 20 = 180 \text{ bytes}$$

Network Layer

Fragment packets when MTU is smaller and assigns offset and Identification fields. Destination host reassembles all fragments into original packet

original packet

12000 Bytes

Router (MTU = 1500)

Destination

- 4) The sliding window protocol is a transport layer mechanism that allows multiple frames to be transmitted before receiving acknowledgements.

Given:

Window size = 6 frames, Total frames = 48, Frame size = 1024

Number of windows

$$48 \div 6 = 8$$

$\therefore$ One frame is lost in every window.

Total retransmissions = 8 frames

- Flow control prevents the sender from overwhelming the receiver, improving efficiency and reducing congestion.
- Sliding window protocol increases communication efficiency while ensuring reliable transmission.

- 5) The Session Layer establishes, manages and terminates communication sessions. It also inserts synchronization checkpoints so that transmission can resume from the last checkpoint after a failure.

Given:

File size = 600 MB, checkpoint every 60 MB, Failure after 390 MB

Calculation:

Checkpoints created 60, 120, 180, 240, 300, 360

Totally 6 checkpoints

Last checkpoint before failure = 360 MB

and to transmit: $390 - 360 = 30 \text{ MB}$

60 - 120 - 180 - 240 - 300 - 360

Restart from 360 MB

Synchronization reduces transmission time and improves communication reliability

The Presentation Layer performs data compression, encryption and format conversion. Compression reduces transmission size, while encryption protects the data from unauthorized access

Given: Original file = 900 MB, Compression = 40%, Encryption = 5%

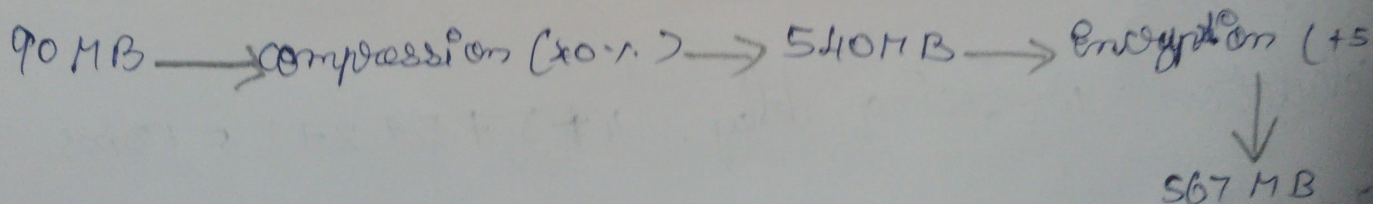
$$\begin{aligned}\text{compressed file size} &= 900 \times 60 \\ &= 540 \text{ MB}\end{aligned}$$

$$\begin{aligned}\text{Encrypted size} &= 540 \times 105\% \\ &= 567 \text{ MB}\end{aligned}$$

$$\begin{aligned}\text{Reduction} &= 900 - 567 \\ &= 333 \text{ MB}\end{aligned}$$

$$\text{Percentage Reduction} = \frac{333}{900} \times 100 = 37\%$$

→ Presentation Layer Improves Both security and transmission efficiency



D) The application layer provides services as FTP, HTTP, SMTP, DNS. It acts as the Interface between users and network application.

Given:

Data = 3 GiB, Throughput = 120 Mbps, Request size = 3 MB

Calculation:

convert: 3 GiB = 3000 MB

Requests: $3000 \div 3 = 1000$ Requests

Transmission time: 3000 = 24000 Mb

$24000 \div 120 = 200$ Seconds

User $\rightarrow$ FTP application $\rightarrow$ 1000 Requests $\rightarrow$ Network $\rightarrow$ Destination.
The Application layer ensures reliable file transfer through user-friendly protocols like FTP.

Each OSI layer contributes a small processing delay. The total communication delay is the sum of processing, transmission and propagation delays.

Given: application = 4 ms, Transport = 3 ms, Network = 5 ms, Data Link = 2 ms, Physical = 1 ms, Transmission = 12 ms, Propagation = 18 ms

Calculation: Processing delay: $4 + 3 + 5 + 2 + 1 = 15$ ms

Total delay: $15 + 12 + 18 = 45$ ms

STP
while total delay determines network performance

Conclusion: Total end-to-end delay = 45ms

Each OSI layer contributes to reliable communication,

The Protocol overhead consists of headers and trailers added by different OSI layers. Although overhead consumes bandwidth, it is essential for addressing, routing and error detection.

Given: Useful data = 8MB, Frame size = 1600 Bytes, overhead = 80 Bytes/frame

Calculation:

Useful payload: $1600 - 80 = 1520$ Bytes

Frames: $80,000,000 \div 1520 \approx 52,632$ Frames

Total overhead: $52,632 \times 80$

$= 4,210,560$ Bytes ≈ 4.21 MB

Efficiency: $\frac{1520}{1600} \times 100 = 95\%$

Reducing protocol overhead improves bandwidth utilization while maintaining reliable communication.

b) Multiple OSI layers work together to provide secure, reliable and efficient communication. The presentation layer

compresses data, the transport layer segments it, the Data Link layer adds frame overhead, and the Physical layer transmits final bit stream

calculation:

compressed file: $240 \times 70\% = 168 \text{ MB}$

convert: $168 \text{ MB} = 168,000,000 \text{ Bytes}$

Segments: $168,000,000 \div 1400 = 120,000 \text{ Segments}$

Protocol overhead: $120,000 \times 60 = 7,200,000 \text{ Bytes} = 7.2 \text{ MB}$

Final transmitted data: $168 + 7.2 = 175.2 \text{ MB}$

240 MB $\rightarrow$ Presentation compression $\rightarrow$ 168 MB $\rightarrow$ Transport
120000 Segments

Physical layer Transmission $\leftarrow$ 175.2 MB $\leftarrow$ Data Link +
B/frame

Conclusion: The cooperation of multiple OSI layers ensures secure, reliable and efficient communication while minimizing transmission size